

10 — Mock paper (MTE syllabus)

Covers the professor's MTE syllabus (10/9/26), Parts I–III · all 14 of the professor's board questions + textbook problems for syllabus items the class has not yet worked · marks unknown (no hand-out or past paper yet) · attempt closed-book, then mark with the answers

HOW TO USE THIS

Do it in one sitting without opening 01–09. Write every step: the working earns marks, not just the final number. Mark each item clean / with struggle / wrong, and schedule it as in 00. Items tagged **textbook** are not the professor's; everything else is quoted from the board.

Section A — Theory

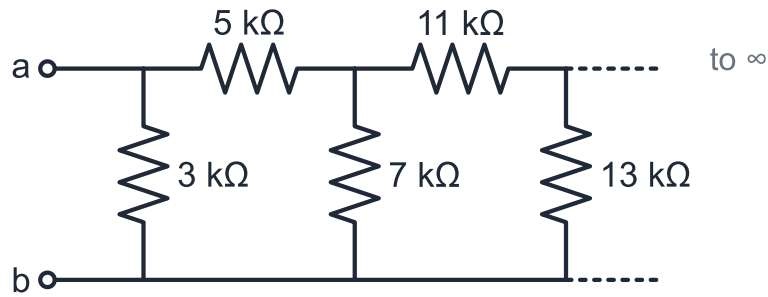
Prompts on the syllabus topics. No past paper has been seen yet, so the *wording* of these is not the professor's.

1. Classify circuit elements. Distinguish active and passive elements, with examples. Draw the symbols of the four dependent sources and name a device for each.
2. Distinguish lumped and distributed parameters.
3. Derive $v = L \, di/dt$ from $\psi = N\phi$. Explain why an inductor does not allow a sudden change of current.
4. State KCL and KVL. State the passive sign convention for power.
5. Write the procedures for nodal and mesh analysis. When is each preferred?
6. State the superposition theorem. List its limitations. How is it applied when sources have different frequencies?
7. Write the procedure for Thévenin's theorem. How is R_{Th} found when the network contains dependent sources?
8. State Norton's theorem. Give its procedure and draw the equivalent model.
9. Derive the condition for maximum power transfer to a resistive load. State the condition for a complex load.
10. Explain $t = 0^-$, 0^+ and ∞ . Write the general first-order response and the time constants of RL and RC circuits.
11. Define the unit impulse, step and ramp signals, with their Laplace transforms and one example each.
12. For a series RLC circuit, write the characteristic equation and classify the response by damping.

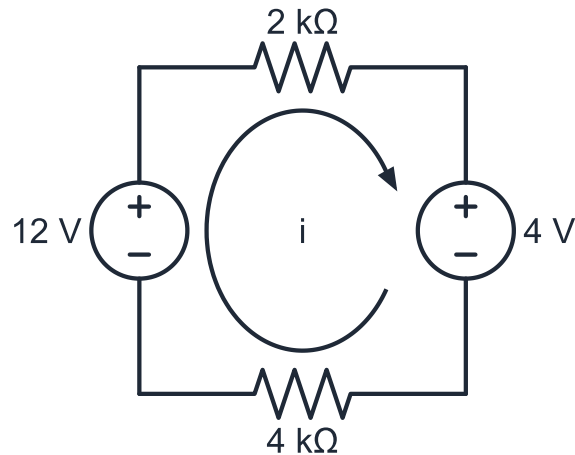
Section B — Numericals

Part I · elements, laws, analysis

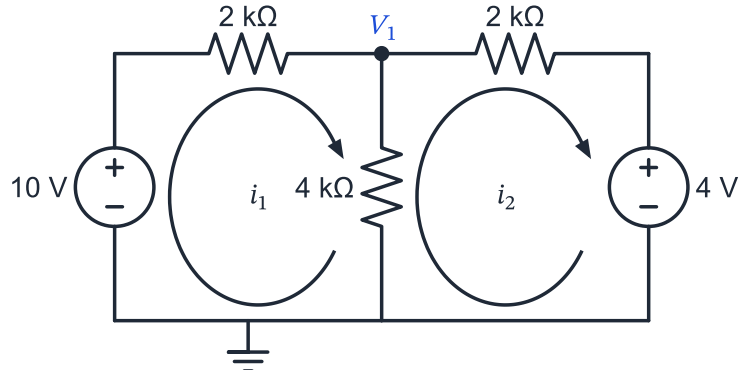
- B1** **Class p.4** 15 V is applied across 3 k Ω and 6 k Ω in series. Find V_1 and V_2 .
- B2** **Class p.5** A 12 A current source feeds 3 k Ω , 6 k Ω and 4 k Ω in parallel. Find I_1 , I_2 , I_3 .
- B3** **Class p.5** $v = 10 \sin(2000t + 30^\circ)$ V across 3 k Ω and 7 k Ω in series. Find v_1 , v_2 .
- B4** **Class p.1** Find R_{eq} of the infinite ladder 3 k Ω shunt, 5 k Ω series, 7 k Ω shunt, 11 k Ω series, 13 k Ω shunt, ...



B5 [textbook](#) Find the loop current and the power of every element. Show that the powers balance.

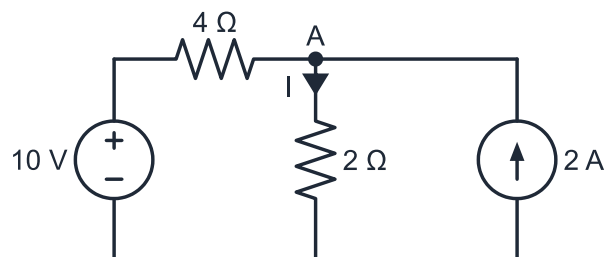


B6 [textbook](#) Find V_1 by nodal analysis and the mesh currents i_1 , i_2 by mesh analysis. Check that they agree.

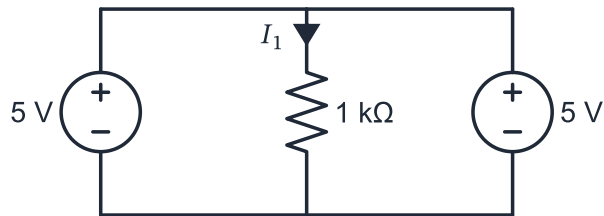


Part II · theorems (DC and AC)

B7 [Class p.7 · set 2 p.8](#) Calculate the current I using superposition. Verify by nodal analysis.

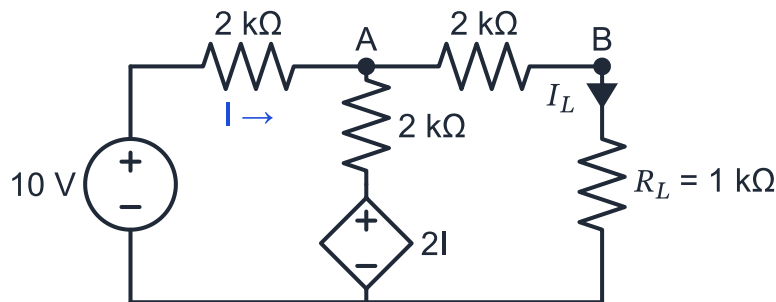


B8 [Class p.9 · set 2 p.9](#) Calculate the current I_1 using superposition. Comment on the result.



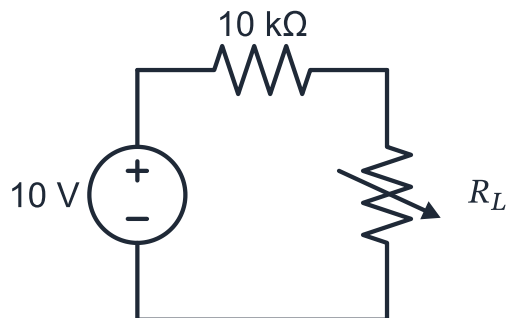
B9 [textbook](#) A 10 V DC source and a $10 \cos(1000t)$ V source in series drive $1 \text{ k}\Omega$ and 1 H . Find $i(t)$ in steady state.

B10 [Class p.11 · set 2 p.11](#) Determine the load current using Thévenin's theorem (take $2I$ as $2 \text{ k}\Omega \times I$, I in mA). Verify with Norton's theorem.



B11 [textbook](#) $10\angle 0^\circ \text{ V}$ at $\omega = 1000 \text{ rad/s}$, $1 \text{ k}\Omega$ in series, $1 \mu\text{F}$ across the output. Find the Thévenin and Norton equivalents, the load for maximum power, and that power.

B12–B13 [Class p.16–17 · set 2 p.14](#) For the circuit below: (a) the value of R_L for maximum power; (b) the total power delivered by the source; (c) the power consumed by R_{Th} ; (d) the efficiency; (e) the voltage across the load; (f) the relation between load power and total power, and between load voltage and source voltage.



B14 [Class p.17](#) 10 V source with $5 \text{ k}\Omega$: determine R for maximum power, the maximum power, and the efficiency.

Part III · transients

B15 [Class p.19–20](#) V , switch closing at $t = 0$, R . Find $I(0^-)$, $I(0^+)$, $I(\infty)$.

B16 [Class p.21](#) V , switch closing at $t = 0$, R and L in series. Find I_R , I_L at 0^- , 0^+ , ∞ . Then find $i(t)$ for $V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$, **by Laplace transform**.

B17 [Set 2 p.20–21](#) V , switch closing at $t = 0$, R and C in series (C uncharged). Find v_C , v_R , I at 0^- , 0^+ , ∞ .

B18 **Class p.22** 10 V, a switch opening at $t = 0$, $1\text{ k}\Omega$ and 1 mH in series. Find $I_L(0^-)$ and $I_L(0^+)$. What happens to the inductor's current after the switch opens?

B19 **textbook** An RC circuit ($R = 1\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$, output v_C) is driven by a unit step, then by a unit ramp. Find $v_C(t)$ for each by Laplace transform.

B20 **textbook** 10 V is switched onto a series R, 1 H , $1\text{ }\mu\text{F}$ circuit (at rest). Classify the response for $R = 2.5\text{ k}\Omega$, $2\text{ k}\Omega$, $1.2\text{ k}\Omega$, 0 , and find $v_C(t)$ for $R = 2.5\text{ k}\Omega$.

Answers

Section A: model answers are the **Exam form** sections: 01 (A1–A3), 02–03 (A4), 04 (A5), 05 (A6), 06 (A7, A8), 07 (A9), 08 (A10, A11), 09 (A12).

	Answer	Where it's taught
B1	$V_1 = 5 \text{ V}, V_2 = 10 \text{ V}$	02 §1
B2	5.33 A, 2.67 A, 4 A	02 §2
B3	$v_1 = 3 \sin(2000t + 30^\circ) \text{ V}, v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$	02 §3
B4	$\approx 2.318 \text{ k}\Omega$ (if the values continue as primes)	02 §5
B5	$i = 4/3 \text{ mA}$; 12 V source -16 mW (delivers), 4 V source $+5.33 \text{ mW}$, $2 \text{ k}\Omega + 3.56 \text{ mW}$, $4 \text{ k}\Omega + 7.11 \text{ mW}$; sum 0	03 §2–3
B6	$V_1 = 5.6 \text{ V}$; $i_1 = 2.2 \text{ mA}$, $i_2 = 0.8 \text{ mA}$; $4 \text{ k}\Omega$ carries $1.4 \text{ mA} = 5.6 \text{ V}/4 \text{ k}\Omega \checkmark$	04 §1, §3
B7	$I_1 = 5/3 \text{ A}$, $I_2 = 4/3 \text{ A}$, $I = 3 \text{ A}$ (resistors in Ω); nodal $V_A = 6 \text{ V}$	05 §1–3, 04 §2
B8	By the rule each case gives 0 (the shorted idle source also shorts the $1 \text{ k}\Omega$), so $I = 0 \text{ A}$; true value 5 mA . Limitation: superposition fails for parallel ideal voltage sources	05 §5
B9	$i(t) = 10 + 7.07 \cos(1000t - 45^\circ) \text{ mA}$	05 §6
B10	$V_{Th} = 20/3 \text{ V}$, $R_{Th} = 8/3 \text{ k}\Omega$, $I_L = 20/11 \approx 1.82 \text{ mA}$; Norton $I_N = 2.5 \text{ mA}$, same I_L . (With $2I$ read as 2Ω : 5.0025 V , $3.0 \text{ k}\Omega$, 1.25 mA ; see 06 §2 Note)	06 §2–5
B11	$V_{Th} = 7.07 \angle -45^\circ \text{ V}$, $Z_{Th} = 500 - j500 \Omega$, $I_N = 10 \angle 0^\circ \text{ mA}$; $Z_L = 500 + j500 \Omega$, $P_{max} = 12.5 \text{ mW}$	06 §6, 07 §4
B12–B13	(a) $10 \text{ k}\Omega$ (b) 5 mW (c) 2.5 mW (d) 50% (e) 5 V (f) $P_L = P_s/2$, $V_L = V_s/2$	07 §2–3
B14	$R = 5 \text{ k}\Omega$, $P_{max} = 5 \text{ mW}$, $\eta = 50 \%$	07 Check 1
B15	0, V/R , V/R	08 §1
B16	0, 0, V/R for both; $I(s) = 10^4/(s(s + 10^6)) \Rightarrow i(t) = 10(1 - e^{-t/1 \mu s}) \text{ mA}$	08 §2, §6
B17	0^- : all 0. 0^+ : $v_C = 0$, $v_R = V$, $I = V/R$. ∞ : $v_C = V$, $v_R = 0$, $I = 0$	08 §2
B18	$I_L(0^-) = I_L(0^+) = 10 \text{ mA}$. With no path, the ideal circuit has no solution (voltage spike, arc). If the R–L loop closes on itself: $i(t) = 10e^{-t/1 \mu s} \text{ mA}$	08 §5
B19	$\tau = 1 \text{ ms}$. Step: $1 - e^{-t/\tau} \text{ V}$. Ramp: $t - \tau + \tau e^{-t/\tau}$ (lags the input by τ)	08 §7
B20	$\omega_0 = 1000 \text{ rad/s}$, $R_c = 2 \text{ k}\Omega$: over, critical, under, undamped. $R = 2.5 \text{ k}\Omega$: $v_C = 10 - (40/3)e^{-500t} + (10/3)e^{-2000t} \text{ V}$	09 §4–5